

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 215 — Past Questions & Materials
by Renance

Course: MEE 215
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — MEE 215 (2015 /2016)

Level 200 • First Semester • PastQuestion

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First, Set: £: Bc] 2015/2016 Academie Sessio:: 4

COURSE CC: Es : Lé: MEE 215 - FLUID MECHANZes 3, hae

Ex: DATEç 025. 2016" -TEME ALLOWEDY2 Hrs~

INSTRUCTIONS: "ANS 27 FSTEON ONE: AND ANY OTHER THRE

pCa Gren 'ae ie a i

ITION 1 (COMPULSORY) aot ee a bh

My To which states of mew es 2. 0 'cs belong? - -, » a [1 Mark}

ii) Sketch curves of shear <* ç::, 7, versus seis gradient, § =, @, for (a) News: -an Fluids, and :

(b) Dilatants. a 4 {1 Marky *

(iii}% A plate (2m x 2m }, 0.2E ar distant apart from-a fixed plate, moves a> -:C am/s" and requires : B

force of 1N, Determine =e çvinamic viscosity of the fluid fn between the aces. [3 Marks

(iv). A crude carrier of weight 2.4 kN is made to fléat on-a padded wooden mer2" fal of weight, 600 Ne

for its movement in sea, Given that the density of sea water-is 1100 ko..n°, what weigat and,

volume o of sea water would be displaced? }- . ° P oe , (4 Marks {

« de é i A

(v) What are the dimensions in terms of My: er, "(Masé: 'Length, Time) oF power and: Kinematic : i _ viscosity? eA , an Marks]

(vi), Given that 8 m? of oil has a gravimetric wert of 1200 Ls Calculate its mass' Gensitys specific.

; - weight, specific gravity and Specific volume: * [4 Marks}

(vii) Determine (a) the gauge pressure and (b).the abet pebine of water ai a depth of 9: from J the surface. ~ . | : [2 Marks]

viii) - With the aid of mathematical. expressions differentiate between he: following types of Flow: (i)

Us ae Steady: and unsteady flow (ii) uniform and ion uniform flow ii) compressiPle and incompressible

ad Flow ss A preity ier a fic ho

(ix) Oil, density p=-727 kg/m, dynamic viscosity, b= 4.94 x 10% 'tow wit th Velocity, Ve 1.39 m/s i in'a

_ pipe of diameter, BD = 0,5 m, Calculate toe aprile number. State with reason whether! the flow a an is laminar or turbulent, 2s sie ds te eee be : a [3 iMarks]

ol ~ Mention the two major approaches 'that. joe been developed. for desejib per the motion of flow'

sohachrlesd in amoving Phd. iff erentiate between them. oe ay A pe. "TR Marks,

questzonzr2) ©. ! Se Ge SM Oe fee fw

4

(a) The radius: $R =$ at time t , of the fireball of a nuclear blast-which occurs at $t = 0$, with energy release E , in air of density ρ , is given by: $R = \sqrt[5]{E t^2 / \rho}$. Determine whether or not the equation satisfies Dimensional Homogeneity (5 Marks)

(b) In a certain flow experiment, the drag force F is thought to depend on the diameter D , the fluid viscosity, μ , and density, ρ . Using the Buckingham's Pi theorem, determine the number of Pi groups and:

olve the non-dimensional Pi groups, $\pi_1 = \dots$

QUESTION 3, e, : | ae

(a): A submarine 156 m long is to move at a velocity of 683. m/s on the surface of the sea. It should a geometrically similar model 2.44 m long be tested?

¢ Oo! G jis: HUT oe a gSatts pee

' Persrex winds Ts orihe vg fica ie

(ey A circular 1m diameter yt TSO ol Detenniine the magni ftide of

ter.sur

water (Fig Q3a). The top of the window is 2m below the water

4, E the total resultant force acting on the window, and the location of the centre of pressure. Take the specific weight of sea water as 9.81 kN/m³

P Y state Pascal's law, for pressure at a point. (5 Marks)

oe 6) A U-tube manometer (Fig, Q4b) measures the pressure difference between two points A and B in a liquid of density ρ . Calculate the difference of

pressure of density ρ . The U-tube contains mercury

of density ρ_m . The U-tube contains mercury

find the value of a if $a = 1.5$ m, $b = 0.75$ m and $\rho_m = 13600$ kg/m³, given that the liquid at A and B is water and $\rho = 1000$ kg/m³. (12 Marks)

QUESTION 5, f aie er : ae

(a) The three dimensional incompressible flow is defined by $v = ax + by + cz$ and $w = bx^2 - cy^2 + az^2$, distribution such that

of, the continuity equation is satisfied, (5 Marks)

ible flow, the velocity components in y and z

Determine the third component of velocity

a

v-

diameters

200 mm and 100 mm: at sections A, and B

respectively. The rate of flow through the pipe is 35 litres/s. Section A is 6 m above datum and

section B is 4 m above datum, If the pressure at section A is 400 kN/m², find the intensity of head losses through the pipe are infinitesimally small.

ies yes is flowing through a pipe having:

oe at section B, such that the frictiona' é

es a bg: ; al [9 Marke]

QUESTION 6 Be os eee vias

é (a) Mention any © ther two 9 (2) flow measuring devices, ather t+han the venturi meter, [2 Marks] i

ha b) A venturi meter having a throat 'diameter d2-of 100 mm is fitted into a pipeline which has a diameter d

3 yy = dil 'of 250 mm through 1 which oil of specific gravity 0.9 is flowing. The pressure difference between

Bf oi ft. sthe entry and throat tapping is measured by a U-tube' manometer; -containing_mercury, 'of. specific

gravity 13,6, and the connections are filled with the oil Flows in the pipeline, If the difference o

: yiridicated pine mercury in the U-tube is 0.63 m, calculate: the. theoretical volume Pate oF Fp

bho ee [6 Marks

"ASA jer engine, consumes: 20.67. bales af fuel andj jrgests 100 ita T air 'in flight: at 900.km/hr in stillai.:

~The exhaust gases are expelled at 700, Tvs relative to the-engine:. Thé static pressures ofid is ,

"cross: éctiohal- "areas at the inlet and outlet af the | engine are equal.' 'Deter mine:the thrust, of;

; ae 36° Marks

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. wee = L H eee ® : ? 7 |

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Paper 2 of 3 — MEE 215 PAST QUESTION

Level 200 • First Semester • PastQuestion

MEE 215 Problem Set 1 – Due for Submission 3 pm on Thursday, 22/02/2024

1a. Calculate the specific weight w , the density ρ and the specific gravity for oil of which 10 m³ weighs 78.33 kN. Assume g , the acceleration due to gravity is 9.81 m/s², and the specific weight of water is 9.79 kN/m³.

1b. A liquid has a mass density of 1550 kg/m³. Calculate its specific weight, specific gravity, and specific volume.

2a. A plate (2m x 2m), 0.25 mm distant apart from a fixed plate, moves at 40 cm/s and requires a force of 1 N. Determine the dynamic viscosity of the fluid in between the plates.

2b. At a certain point in an oil the shear stress is 0.2 N/m² and the velocity gradient is 0.21 s⁻¹. If the mass density of the oil is 950 kg/m³ find the kinematic viscosity.

3a. Determine the absolute pressure of oil at the point in its storage vessel when gauge pressure reads 80 kPa.

3b. Convert a pressure head of 10 m of water column to kerosene of specific gravity 0.8 and carbon-tetra-chloride of specific gravity of 1.62.

3c. Determine (a) the gauge pressure and (b) The absolute pressure of water at a depth of 9 m from the surface.

3d. A U-tube manometer (illustrated below) measures the pressure difference between two points A and B in a liquid of density ρ_1 . The U-tube contains mercury of density ρ_2 . Given that $a = 1.5$ m, $b = 0.75$ m and $h = 0.5$ m, solve for the difference of pressure in pipes A and B if they both contain water, and given that the specific gravity of mercury is 13.6.

4a. An Isosceles triangular plate of base 5 m and altitude 5 m is immersed vertically in an oil of specific gravity 0.8. If the centre of gravity of the triangle is 2.67 m below the free water surface, determine the total pressure and the centre of pressure. Take specific weight of water as 9.79 kN/m³.

4b. A wooden block of width 2.5 m and length 6 m is floating horizontally in water. If the wooden block is 0.97 m deep into the water, find (i) the volume of water displaced, (ii) position of centre of buoyancy and (iii) weight of the block. Take specific weight of water as 9.79 kN/m³.

Paper 3 of 3 — MEE 215 Handout

Level 200 • First Semester • Handout

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 2

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
- Concepts of velocity distribution, pressure drop, and shear stress in pipe flows
- Fundamentals of dimensional analysis; analysis of selected dimensionless parameters and their significance (Reynolds number as an example)
- Introduction to flow measurement – concepts and measuring devices

2

Fluid Properties

> Newton's Law of Viscosity

The shear stress (τ) acting on a fluid element layer is directly proportional to the velocity gradient. The constant of proportionality is called the co-efficient of viscosity or dynamic viscosity (μ). If divided by density, kinematic viscosity (ν) is obtained.

□□

It states mathematically that , □□ being the rate of shear strain, or the velocity gradient induced by the action of the shear stress on the fluid element. Fluids following the Newton's law of viscosity are known as Newtonian fluids, such as water, kerosene, and air, while those not following the law are generally categorized as non-Newtonian fluids, such as blood, mud flows, polymer solutions, slurries, etc. Several other types of fluids classify as non-Newtonian, such as ideal fluids, plastic fluids (real plastics and ideal/Bingham plastics), dilatent fluids, thixotropic

substances. The following graph defines the stress-velocity gradient relationship for the different types of fluids.

3

Fluid Properties

> Newton's Law of Viscosity

The mathematical expressions for some of the fluid types are:

In case of non-Newtonian fluids, if n is less than unity they are called pseudo-plastics (e.g., paper pulp, rubber suspension paints) while fluids in which n is greater than unity are known as dilatents. (e.g., Butter, printing ink).

Exercise - mention at least 3 examples of other types of non-Newtonian fluids. 4

Fluid Properties

> Newton's Law of Viscosity - CPQs

1. Consider a flat metal strip placed in parallel at the top of a fixed plate at a distance of about 0.000075 m, with hydraulic oil ($\rho = 869 \text{ kg/m}^3$) in between them. Given that the top plate is sheared and kept in motion at a velocity of 1.5 m/s by a force 2.5 N. Solve for the dynamic and kinematic viscosities of the hydraulic oil.

2. Given that a flow over a plate is characterized by the velocity distribution, $u = u_{\max} - \frac{y^2}{2\delta^2} u_{\max}$, where u is the velocity in m/s of the flow at a distance y m above the plate. Determine the flow velocity gradient and the shear stress right on the surface of the plate and at a height 1.8 m above it. Assume 0.9 Pa·s for the dynamic viscosity of the fluid.

5

Fluid Statics, Kinematics, and Dynamics

> Overview of Fluid Statics, Kinematics, and Dynamics

✓ Fluid Statics

- It deals with problems associated with fluids at rest. There is no relative motion between adjacent fluid layers, and thus there are no shear (tangential) stresses in the fluid trying to deform it.
- The only stress in consideration in fluid statics is the normal stress, which is the pressure, and the variation of pressure is due only to the weight of the fluid.
- Thus, the topic of fluid statics has significance only in gravity fields, and it concerns principally the study of pressure (normal forces) and its effects on engineering bodies.
- The design of many engineering systems such as water dams and liquid

storage tanks requires the determination of the forces acting on their surfaces using fluid statics.

- Fluid statics is generally referred to as hydrostatics when the fluid is a liquid and as aerostatics when the fluid is a gas.

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Fluid Statics, Kinematics, and Dynamics

> Overview of Fluid Statics, Kinematics, and Dynamics

✓ Fluid Kinematics

- It deals with the study of how fluid flows, that is, the velocity and acceleration of particles of fluid in motion and their distribution in space without considering any force or energy involved.
- The motion of fluid particles can be described by using either the Lagrangian method or the Eulerian method.

✓ Fluid Dynamics

- Like fluid kinematics, fluid dynamics also deals with the study of flow of liquids and gases. Thus, fluid kinematics can be said to be a branch of fluid dynamics.
- However, unlike in fluid kinematics where forces and energy causing the flow are not considered, fluid dynamics studies the flow of fluids under the influence of forces and energy.

More attention is given here to fluid statics as described in the following slides, but very basic concepts of fluid kinematics and dynamics are also defined herein and in later topics under conservation laws and equations. 7

Fluid Statics

> Pressure

✓ Definition

- When a fluid is contained in a vessel, it exerts force at all points on the sides and bottom and top of the container. The force per unit area is called pressure.

□

- If, F = the force, and A = Area on which the force acts; then pressure, $p =$

□

- Again, the pressure of a fluid on a surface will always act normal to the surface.
- In a liquid, pressure builds up due to the weight of the fluid, and this pressure increases as the depth of the liquid increases.

For a liquid-filled container shown, with cylinder inserted, the total pressure at the bottom of the pressure would correlate with the height of the cylinder, equal to the weight of the liquid in the cylinder. Mathematically, $p = \frac{F}{A} = \frac{W}{A}$.

8

Fluid Statics

> Governing Principles/Laws

✓ Hydrostatic law

• The rate of increase of pressure in a vertically downward direction must be equal to the specific weight of the fluid at that point, $\frac{dp}{dz} = \rho g$.

✓ Pascal law

• The intensity of pressure at any point in a liquid at rest, is the same in all directions.

> Pressure Forms / Indications

✓ Atmospheric pressure

• It's the normal pressure exerted by the atmospheric air upon all surfaces with which it is in contact, otherwise known as Barometric pressure.

• The atmospheric pressure at sea level (above absolute zero) is called 'Standard atmospheric pressure.'

9

Fluid Statics

> Pressure Forms / Indications

✓ Gauge pressure

• It is the pressure, measured with the help of pressure measuring instrument, in which the atmospheric pressure is taken as datum. The atmospheric pressure on the scale is marked as zero.

• Gauges record pressure above or below the local atmospheric pressure, since they measure the difference in pressure of the liquid to which they are connected and that of surrounding air.

• If the pressure of the liquid is below the local atmospheric pressure, then the gauge is designated as 'vacuum gauge' and the recorded value indicates the amount by which the pressure of the liquid is below local atmospheric pressure, i.e. negative pressure, otherwise known as vacuum pressure.

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Fluid Statics

> Pressure Forms / Indications

✓ Absolute pressure

• It is necessary to establish an absolute pressure scale which is independent of the changes in atmospheric pressure. A pressure of absolute zero can exist only in complete vacuum.

• Any pressure measured above the absolute zero of pressure is termed 'absolute pressure.'

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$$

$$p_{\text{absolute}} = p_{\text{gauge}} + p_{\text{atm}}$$

Pressure is measured fundamentally in Newton per square metre (N/m²), known also as Pascal. Low pressures are often expressed in terms of mm of water or mm of mercury. This is an abbreviated way of saying that the pressure is such that will support a liquid column of stated height. Relationship between pressures

11

Fluid Statics

> Pressure Measurement – achieved using manometers or mechanical gauges

✓ Manometers

- Devices used for measuring the pressure at a point in a fluid by balancing the column of fluid by the same or another column of liquid. These are classified as (a) Simple manometers, such as (i) Piezometer, (ii) U-tube manometer, (iii) Single column manometer, and (b) Differential manometers for measuring pressure difference between two points in a pipe or two pipes, such as (i) U-tube differential manometer, (ii) inverted U-tube differential manometers.

Vertical single column U-tube differential manometer

Piezometer U-tube manometer manometer 12

Fluid Statics

> Pressure Measurement – achieved using manometers or mechanical gauges

✓ Mechanical Gauges

- Devices in which the pressure is measured by balancing the fluid column by spring (elastic element) or dead weight. Generally these gauges are used for measuring high pressure and where high precision is not required. Examples are Bordon tube pressure gauges and Diaphragm gauges.

Bordon tube pressure gauge Diaphragm gauge 13

Fluid Statics

> Examples/CPQs

1. A U-tube manometer is used to measure the pressure of oil of specific weight 8.34 kN/m^3 flowing in a pipe line. Its left end is connected to the pipe and the right limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific weight = 133.42 kN/m^3) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.

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Fluid Statics

> Examples/CPQs

2. A U-tube differential manometer connecting two pressure pipes at A and B is shown below. The pipe A contains a liquid of specific gravity 1.8 under a pressure of 130

□□/□□ . The pipe B contains oil of specific gravity 0.7 under a pressure of 220 □□/□□ .
Find the difference of pressure measured by mercury as fluid filling U-tube.

15

Fluid Statics

> Examples/CPQs

3. A U-tube differential manometer connected at two points A and B is shown below.
Given that air pressure at point A is 140 □□/□□ , solve for the absolute pressure at point B.

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Fluid Statics

> Examples/CPQs

4. An inverted differential manometer is shown below, containing an oil of specific gravity 0.8 connected to two different pipes carrying water under pressure. Given that the pressure in the pipe A is 20 □□/□□ , determine the pressure in pipe B.

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THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lecture 1

Course: Fluid Mechanics I – MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

1

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✓ Basic Concepts

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- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
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- Introduction to flow measurement – concepts and measuring devices

2

Learning Objectives (Part 1)

- To define and characterize fluid/flows and their properties, and fluid mechanics.
- To differentiate amongst fluid mechanics, fluid kinematics, and fluid dynamics, and apply classical models on each of them to solve simple engineering problems.
- To identify the importance of control system/volume definition in thermo-fluid problem-solving and define the same for solving specific simple problems.
- To state the classical continuity, momentum, and energy equations (models) and apply the same to solve simple engineering problems.

N.B – Course Learning Outcomes (CLOs) are tied to the learning objectives, with students expected to be able to define and characterize flows and minimum of 5 properties (density, specific volume, specific weight, viscosity, specific heat capacity, etc) in objective 1 above – similar CLOs apply for all the other learning objectives.

3

Course Organization

A 2 credit unit course (2 hours of lectures/recitations per week for 15 weeks).

CA (30 marks) will comprise CPQs/quizzes and problem sets (assignments). Final exam (70 marks). Letter grades as defined in the college handbook.

Flipped classroom and peer instruction teaching/learning approach would be adopted. CPQs could be at the beginning of a class, to assess your participation in the out-of-class activities.

Communications to be handled online via piazza.

Some good textbooks consulted while preparing this and subsequent lecture slides (intellectual credit to authors of these texts):

- ✓ Fluid Mechanics: Fundamentals and Applications, 4th edition by Yunus A. Çengel and John M. Cimbala.
- ✓ A Textbook of Fluid Mechanics and Hydraulic Machines by R. K. Rajput

4

Course Outline

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5

General Introduction

How does a wind power plant work? What basic components and thermo-fluid processes are involved?

6

Introduction - Definitions

> What is fluid mechanics?

What is a fluid? – A substance that flows.

What is mechanics? – A branch of science dealing with the study of objects/processes under the influence of force. Mechanics could be categorized as static (concerning stationary body) or dynamic (relating to body in motion).

Then, what is fluid mechanics?

Fluid mechanics is the science that deals with the behavior of fluid at rest (fluid statics) or in motion (fluid dynamics), and the interaction of fluid with solids or other fluid at the boundaries.

7

Introduction - Definitions

> What is fluid mechanics?

Distinction between a solid and a fluid is made on the basis of the substance's ability to resist an applied shear (or tangential) stress that tends to change its shape.

Stress is defined as force per unit area and is determined by dividing the force by the area upon which it acts.

The normal component of a force acting on a surface per unit area is called the normal stress, and the tangential component of a force acting on a surface per unit area is called shear stress.

In a fluid at rest, the normal stress is called pressure. A fluid at rest is at a state of zero shear stress.

8

Properties of Fluid

> Density

Mass density (or specific mass) of a liquid is defined as mass per unit volume at standard temperature and pressure.

Weight density (or specific weight) is defined as weight per unit volume at

standard temperature and pressure, denoted by w , equal to g (9.81 for water).

Specific volume (v) is defined as volume per unit mass at standard temperature and pressure.

What's the relationship between density and specific volume?

9

Properties of Fluid

> Specific gravity

Specific gravity of a fluid is defined as the ratio of its specific weight to that of a standard reference fluid. For liquid, the standard reference fluid is water at 4 degrees C.

> Viscosity

It may be defined as the property of a fluid which determines its resistance to shearing stresses (resistance to flows). It's responsible for the no-slip condition.

What is no-slip condition?

> Example

Calculate the specific weight, density, specific volume, and specific gravity of a liquid having a volume of 8 m³ and weight of 48 kN.

10

Classification of fluid flows

> Classification based on the mode of initiation of fluid motion

Natural (free) flow, when fluid motion is Forced flow, when fluid motion is due to natural means such as the compelled to flow over a surface or in a buoyancy effect; which manifests itself as pipe by external means such as a pump or the rise and fall of warmer (lighter) and a fan. colder (denser) fluids.

> Classification based on variability of flow properties with time

Steady flow, where fluid property at a Unsteady flow, where fluid property at a given location does not change with time. location changes with time.

It's different from uniform flow, where a

Turbines, heat exchangers, condensers

given flow property is the same at all

and pumps work for a long time under the

locations at a given time.

same conditions, and are steady-flow 11

Classification of fluid flows

> Classification based on the nature of boundary of flow

Internal flow, when fluid flows in a pipe or External flow, when unbounded fluid flows duct completely bounded by solid over a surface such as a plate, a wire or a

surfaces. E.g. Flow of crude oil in a pipe. pipe. E.g. Airflow over an exposed pipe
 Flow of a liquid in duct that is partly-filled during a windy day.
 is called open-channel flow; such as flow
 of water in a river.

N.B. Internal flows are dominated by the influence of viscosity throughout the flow field.
 In external flows, viscous effects are limited to boundary layers near solid surfaces and
 to wake regions downstreams of bodies.

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Classification of fluid flows

> Classification based on the level of variation of density during flow

Compressible flow, the case that density of fluid changes during flow. Incompressible flow, the case that
 density of fluid remains almost constant throughout the flow. The volume of every
 quantity of fluid, the volume changes throughout the flow. The volume of every
 considerably at different sections of flow. portion of fluid remains nearly the same
 It applies mostly to gases, although not over the course of its motion. Flows of
 all flows of all gases can be considered almost all liquids can be considered
 compressible. incompressible to a very high degree.

Flows of some gases can also be
 idealized as incompressible to a
 reasonable level of accuracy, if mach

number, Ma

$Ma < 0.3$.

How would you classify the flow of air at 95 m/s on the basis of density variability,
 considering that the speed of sound in air at room temperature is 346 m/s? Why? 13

Classification of fluid flows

> Classification based on orderliness of fluid molecules during flow

Laminar flow, where the fluid motion is highly ordered, characterized by smooth layers of fluid. It is common with flow of by high velocities and velocity
 Turbulent flow, where the fluid motion is highly disordered/chaotic, characterized
 high-viscosity fluids such as oils at low fluctuations. Example is low-viscosity fluids
 velocities. such as air at high velocities.

A flow that alternates between being laminar and turbulent is called transitional flow.

14

Classification of fluid flows

> Classification based on the degree of internal resistance to flow

Viscous flow region, where effects of 'internal stickiness' of fluid are significant. Inviscid flow region, where viscous forces
 are infinitesimally small compared to the
 N.B. All fluid flows exhibit viscous effects to inertia of pressure forces and can be
 some degree. neglected for ease of analysis. It holds

true in many flows of practical interest,
typically in regions very far from the solid
surface.

The flow of an originally uniform fluid stream over a flat plate, and regions of viscous and
inviscid flows 15

Classification of fluid flows

➤ Classification based on the number of axis (dimension) of velocity variation

One-dimensional, Two-dimensional, Three-dimensional,
when flow velocity when flow velocity when flow velocity
varies along one varies along two varies along the three
primary axis. primary axes. primary axes. $V(x,y,z)$

Velocity gradient Velocity gradient in rectangular
along two along one dimension coordinate and
dimensions are is negligibly small. $V(r,\theta,z)$ in cylindrical
negligibly small. coordinate.

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THANK YOU FOR YOUR ATTENTION!

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Lecture 5

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and their significance (Reynolds number as an example)

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2

Fluid Kinematics

> Fluid Kinematics

✓ Again, fluid kinematics concerns the study of fluid flow and description of fluid motion without any recourse to the forces/energy driving the flow.

✓ Lagrangian and Eulerian methods are the two prominent approaches commonly used to describe fluid motion in fluid kinematics.

✓ Description of fluid motion using the Lagrangian method requires that the position (path taken), velocity and acceleration of each fluid particle be tracked within the flow. Invariably, a mass of fixed identity is followed within the flow, analogous to what obtains in a closed (or control mass) thermodynamic system.

CPQs – What's a thermodynamic system? When is a system said to be closed, open, or isolated?

3

Fluid Kinematics

> Fluid Kinematics

✓ In the Eulerian method, a finite volume, known generally as control volume, is first defined through which fluid flows in and out. The control volume is analogous to the open thermodynamic system, illustrated hereunder.

✓ Then, instead of tracking fluid particles as done in the Lagrangian method, field (point) variables are defined within the control volume as functions of space and time.

✓ The field variable at a particular location at a particular time is the value of the variable for whichever fluid particle happens to occupy that location at that time. Examples are velocity and acceleration fields, defined as vectors below.

A typical control volume

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Fluid Kinematics

> Flow Lines

✓ Fluid particles in motion tend to follow some defined lines, known generally as flow lines.

✓ Prominent among the different types of flow lines are: i) Path Line, ii) Stream Line, iii) Stream Tube, and iv) Streak Line.

✓ Please compare and contrast between the different types of flow lines listed above, with the aid of diagrams.

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Fluid Kinematics

> Examples/CPQs

1. Solve for the velocity and acceleration at a point (2, 4, 6) after 2 seconds in a 3-dimensional flow given by $u = yz + t$, $v = xz - t$, $w = xy$ m/s.
2. Given the velocity field in fluid flow as $V = (6x+4y)i + (4z+6x^2)j + (t-2k)z$. Determine: i) the velocity component u, v and w at any point in the flow field, (ii) the speed at point (1,1,1), and (iii) the speed at time $t = 2s$ at point (0,0,4). Would you say the velocity field is steady or unsteady, uniform or non-uniform, 1-dimensional, 2-dimensional, or 3-dimensional? Justify your answer!

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THANK YOU FOR YOUR ATTENTION!

Any other questions?

Federal University of Petroleum Resources, Effurun,
Nigeria

Mechanical Engineering Department, College of Engineering and Technology

Lectures 3 - 4

Course: Fluid Mechanics I - MEE 215

Course Lecturer:

Joseph OYEKALE (PhD-Doctor Europaeus, MNSE, R.Engr)

2023 - 2024 Academic Session

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Course Outline

✓ Basic Concepts

- Definitions and fluid properties; classification of flows
- Fundamentals of fluid statics, kinematics, and dynamics
- Concepts of system and control volume in thermo-fluid problem solving
- Introduction to continuity, momentum, and energy equations and their applications

✓ Characterisation and Measurement of Flows

- Introduction to incompressible viscous flow of Newtonian fluids in pipes
- Concepts of velocity distribution, pressure drop, and shear stress in pipe flows
- Fundamentals of dimensional analysis; analysis of selected dimensionless parameters and their significance (Reynolds number as an example)
- Introduction to flow measurement – concepts and measuring devices

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Fluid Statics

> Hydrostatic Forces

- ✓ Fluid statics is used also to determine the forces acting on floating or

submerged bodies and the forces developed by devices like hydraulic presses and car jacks, known generally as hydrostatic forces.

✓ The complete description of the resultant hydrostatic force acting on a submerged surface requires the determination of the magnitude, the direction, and the line of action of the force.

✓ Again, application areas include in the determination of forces acting on surfaces in engineering designs involving stationary fluid, such as in dams and fluid storage vessels.

3

Fluid statics

➤ Hydrostatic forces on submerged surfaces

- On a surface submerged in fluid, hydrostatic forces are developed as the static fluid acts on the surface, whether plane or curved, and we often need to determine the magnitude of the force, known as the total pressure, and its point of application, which is called the center of pressure.

- In most cases, the other side of the plate is open to the atmosphere (such as the dry side of a gate), and thus atmospheric pressure acts on both sides

When analyzing hydrostatic forces

of the plate, yielding a zero resultant. on submerged surfaces, the atmospheric pressure can be

- In such cases, it is convenient to subtract subtracted for simplicity when it atmospheric pressure and work with the gauge acts on both sides of the structure. pressure only.

4

Fluid statics

➤ Hydrostatic forces on horizontally submerged plane surfaces

- Take A = Area of the immersed surface,

x = Depth of horizontal surface from the liquid,

and w = Specific weight of the liquid.

- The total pressure on the surface,

P = Weight of the liquid above the immersed surface

= Specific weight of liquid $\times$ volume of liquid

= Specific weight of liquid $\times$ area of surface $\times$ depth of liquid

= wAx

A plane surface immersed horizontally in a liquid

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Fluid statics

> Hydrostatic forces on vertically submerged surfaces

- Consider a plane vertical surface of arbitrary shape

immersed in a liquid, as shown.

- Let, A = total area of the surface,
- G = centre of the area of the surface (centre of gravity),
- h = depth of centre of area,
- h_p = distance of centre of pressure from the free surface of liquid.
- Taking a thin horizontal strip of the surface with width dx and breadth b , it can be shown that the total pressure acting on the surface, $P = wAx$, holds true for all surfaces immersed in liquid, whether plane or curved.

A surface immersed vertically in a liquid

- Also, $P = \rho A h_p$, where ρ = intensity of pressure on the centre of area of the surface.

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Fluid statics

> Determination of Centre of Pressure

- Since pressure tends to increase with depth in a stationary fluid, the total pressure tends to act at a point on a submerged surface below its centre of gravity. This point of action of the resultant forces of the stationary fluid on the submerged surface, the centre of pressure, is usually expressed as a height from the free liquid surface.

- Please kindly review the mathematical proofs on the determination of the centre of pressure, it is defined by

$$h_p = h_c + \frac{I_{Gc}}{Ah_c}$$

 h_c , where I_{Gc} is the moment of inertia of the h_c h_c h_c

immersed surface about the free liquid surface, otherwise

known as the second moment of area, and I_{Gc} is the moment

A surface immersed vertically in a liquid

of inertia of the surface about its centre of gravity.

CPQ - How does $\bar{x}$ relate with $\bar{y}$, mathematically?

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Fluid statics

> Centre of Gravity (G) and moment of inertia (I) of common geometries

Rectangle

Triangle

Circle Trapezium

8

Fluid statics

> Centre of Gravity (G) and moment of inertia (I) of common geometries

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Fluid statics

> Inclined Immersed Surface

$\bar{x}$

$\bar{y}$

• Total pressure, $P = \rho g \bar{h} A$. $\bar{h} = \bar{y} \sin \theta$

$\bar{h}$

$\bar{h}$

$\bar{h} = \bar{h}_1 \cos \theta + \bar{h}_2$

• Centre of pressure, $\bar{h}_c$

$\bar{h}_c$

An inclined surface immersed in a liquid

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Fluid statics

> BUOYANCY AND STABILITY

• It is a common experience that an object feels lighter and weighs less in a liquid than it does in air. These and other observations suggest that a fluid exerts an upward force on a body immersed in it. This force that tends to lift the body is called the buoyant force.

• The buoyant force acting on a body of uniform density immersed in a fluid is equal to the weight of the fluid displaced by the body, and it acts upward through the centroid of the displaced volume. This is known as Archimedes' principle.

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Fluid statics

> Stability of Immersed and Floating Bodies

• An important application of the buoyancy concept is the assessment of the stability of immersed and floating

bodies with no external attachments.

- We use the classic “ball on the floor” analogy to explain the fundamental concepts of stability and instability. As shown, three balls at rest on the floor.
- Case (a) is stable since any small disturbance (someone moves the ball to the right or left) generates a restoring force (due to gravity) that returns it to its initial position.
- Case (b) is neutrally stable because if someone moves the ball to the right or left, it would stay put at its new location. It has no tendency to move back to its original location, nor does it continue to move away. 12

Fluid statics

➤ Stability of Immersed and Floating Bodies

- Case (c) is a situation in which the ball may be at rest at the moment, but any disturbance, even an infinitesimal one, causes the ball to roll off the hill—it does not return to its original position; rather it diverges from it. This situation is unstable.
- What about a case where the ball is on an inclined floor? It is not appropriate to discuss stability for this case since the ball is not in a state of equilibrium.
- In other words, it cannot be at rest and would roll down the hill even without any disturbance.
- For an immersed or floating body in static equilibrium, the weight and the buoyant force acting on the body balance each other, and such bodies are inherently stable in the vertical direction.

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Fluid Statics

➤ Examples/CPQs

1. Consider a rectangular plate 5 metres long and 2 metres wide immersed vertically in water in such a way that its 5 m side lies in parallel to the water surface and is 1 m below it. Solve for (i) the total pressure on the plate and (ii) the point of action of the total pressure.

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Fluid Statics

➤ Examples/CPQs

2. A circular 1m diameter perspex window is on the vertical side of a fish-holding tank containing sea water (see figure below). The top of the window is 2m below the water surface. Determine the magnitude of the total resultant force acting on the window, and the location of the centre of pressure. The moment of inertia of a circle, diameter d , about its centre of gravity axis is $ICG = \pi d^4/64$. For sea water $w = 9.95\text{kN/m}^3$.

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Fluid Statics

> Examples/CPQs

3. Consider a metallic body floating at the mercury-water interface, shown below, such that 25 percent of its volume is submerged in mercury, and the remaining in water. Solve for the weight density of the metallic body.

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Fluid Statics

> Examples/CPQs

4. A boy of weight 400N is floating on a log of wood of weight 600N in sea water. What are the weight and volume of sea water displaced? Take the density of the sea water as 1100kg/m^3 .

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THANK YOU FOR YOUR ATTENTION!

Any other questions?